

Strong Artificial Intelligence

G Concept

Core and Key

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Technical University of Applied Sciences, Würzburg, 15.05.2026

Repetition „Action“

- What is an action?
- Which basic action types do you know?
- What is an action theory?
- Why do individuals start an action?
- What is an action plan and what components has it?

Workshop „Concept“

- What is not a concept?
- What is a concept?
- Find and describe an example of a motion concept.
- Find and describe an example of a building concept.
- Find and describe an example of a time concept.
- Find and describe an example of a space concept.
- Find and describe an example of a thought concept.
- Describe the concept „mother“.

G Concept

- | | |
|------------------------|-------------------------|
| 1. Entity | Atomic |
| 2. Association | First Benefit |
| 3. Association Network | Foundation for Concepts |
| 4. Concept | Schema Carrier |
| 5. Conceptualisation | Learner |
| 6. Materialisation | Actor |
| 7. Conclusion | Core and Key |

1. Entity

Atomic

1. Entity

is

Entity A

delimitable

atomic (undividable)

*What Characterises an Entity
in Principle?*

It is delimitable.

What are Delimiters?

*e. g. surfaces, different
perceptions, ...*

What is a Physical Entity?

*A conglomerate of matter
and waves.*

What is a Mental Entity?

A representation.

1. Entity

Nature

Entity A

physical

mental

virtual



Virtual World (non-being)

defined mentally, phantasy, stories, fakes, ...

1. Entity

representation
(atomic information)

represented entity

primitive entities
physical (static)
object/place/space

Mental Entity

physical entity

mental entity

virtual entity

primitive concepts

name

number

time (dynamic)

role

variable

known in
programming
languages
as data types

core of
programming

1. Entity

primitive concepts

- name: human created phonetic and/or textual pattern used to identify entities (except names) to **move** information among individuals
- number: observed **frequency** of entities with a special concern (formation of quantities)
- time: position of earth rotation relative to sun (base for **dynamic** or time logic)
- role: **action/behavior** expectation with a more complex concept behind (conceptualisation anchor)
- variable: place holder needed for **thinking** and computing
- physical entity: a physical object itself is not a concept; human defined bordered physical areas/volumes as the base for **static** and spatial logic

1. Entity

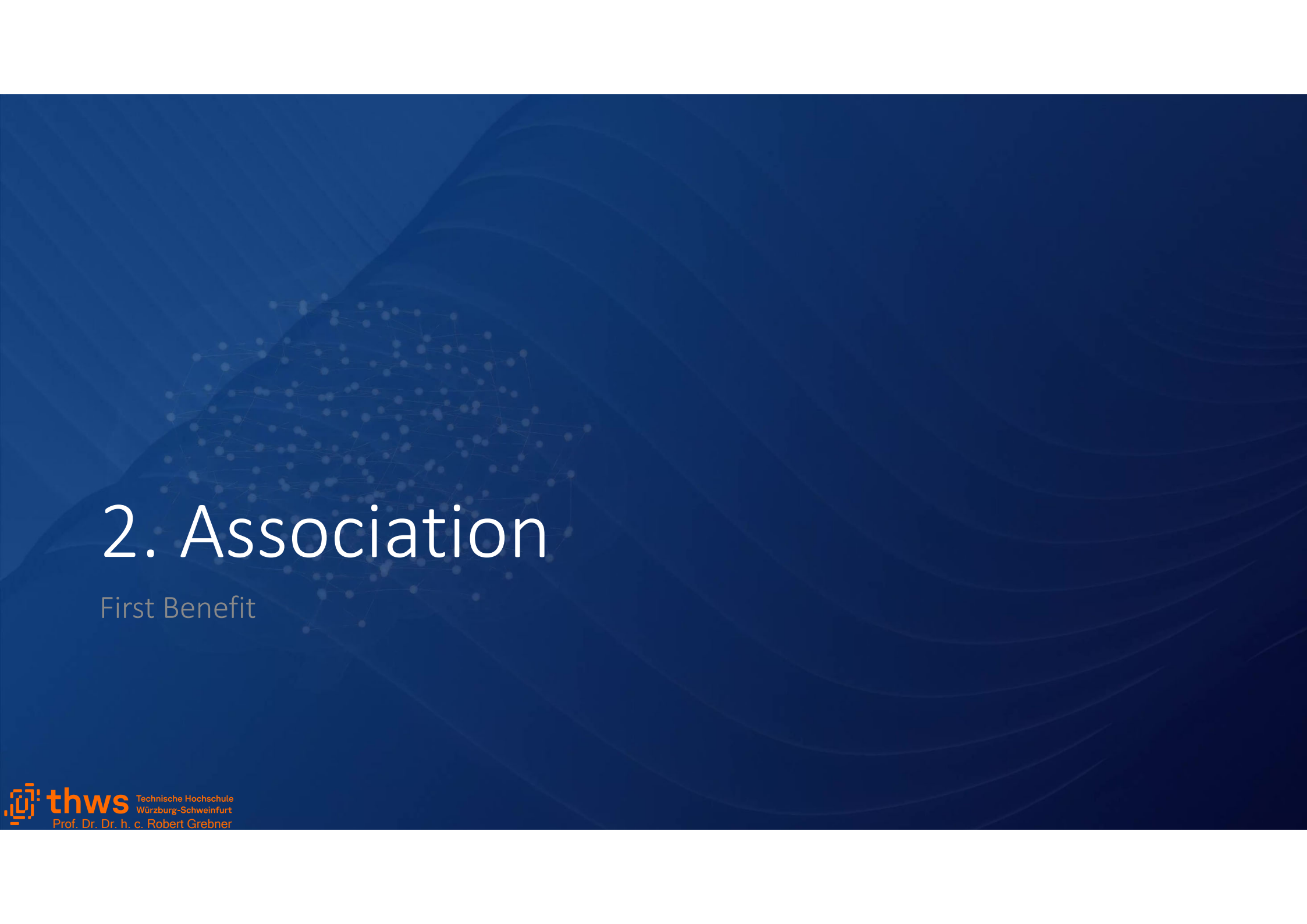
Role as an action or behavior concept

Behind a role there is a concept of an action (active) or behavior (passive).



Where and when do we use variables in thinking or speaking?





2. Association

First Benefit

Workshop „Association“

- What happens very automatically in our brain, when we think about two entities or if we are confronted with two entities at the same time?
- Can we link any arbitrary objects in a meaningful way?
- Why is your brain called to work associative?
- What do you think, if you are confronted with the entity „Cairo Building“ and the entity „WÜbota“?



Cairo Building



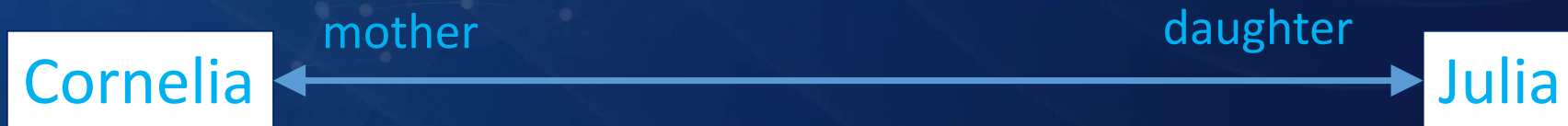
WÜbota

Counterpart paradigm

1. If we link two entities in one association together mentally, we always assign compatible roles to each entity and get a role pair for that association.
2. Roles are compatible, if the role of one entity tells, what that entity is for the other entity and vice versa.
3. Each role represents the action/behavior expectation of its entity
4. The two roles of one association are always counterparts in a sense, that their action or behaviour is usually not contradictory or dead locking – they fit together

2. Association

1. Cornelia **is the mother** of Julia.
2. Julia **is the daughter** of Cornelia.
3. Cornelia and Julia are in a **mother-daughter relation**.
4. Cornelia has the **mother role** in that relation.
5. Julia has the **daughter role** in that relation.
6. The two roles mother and daughter are **counterparts**.
7. The role pair mother&daughter is the **type of the association**, that links Cornelia and Julia.
8. Both roles are coming with compatible **action expectations**.
9. Cornelia can only be called a mother, if she has a daughter.
10. Julia can only be called a daughter, if she has a mother.

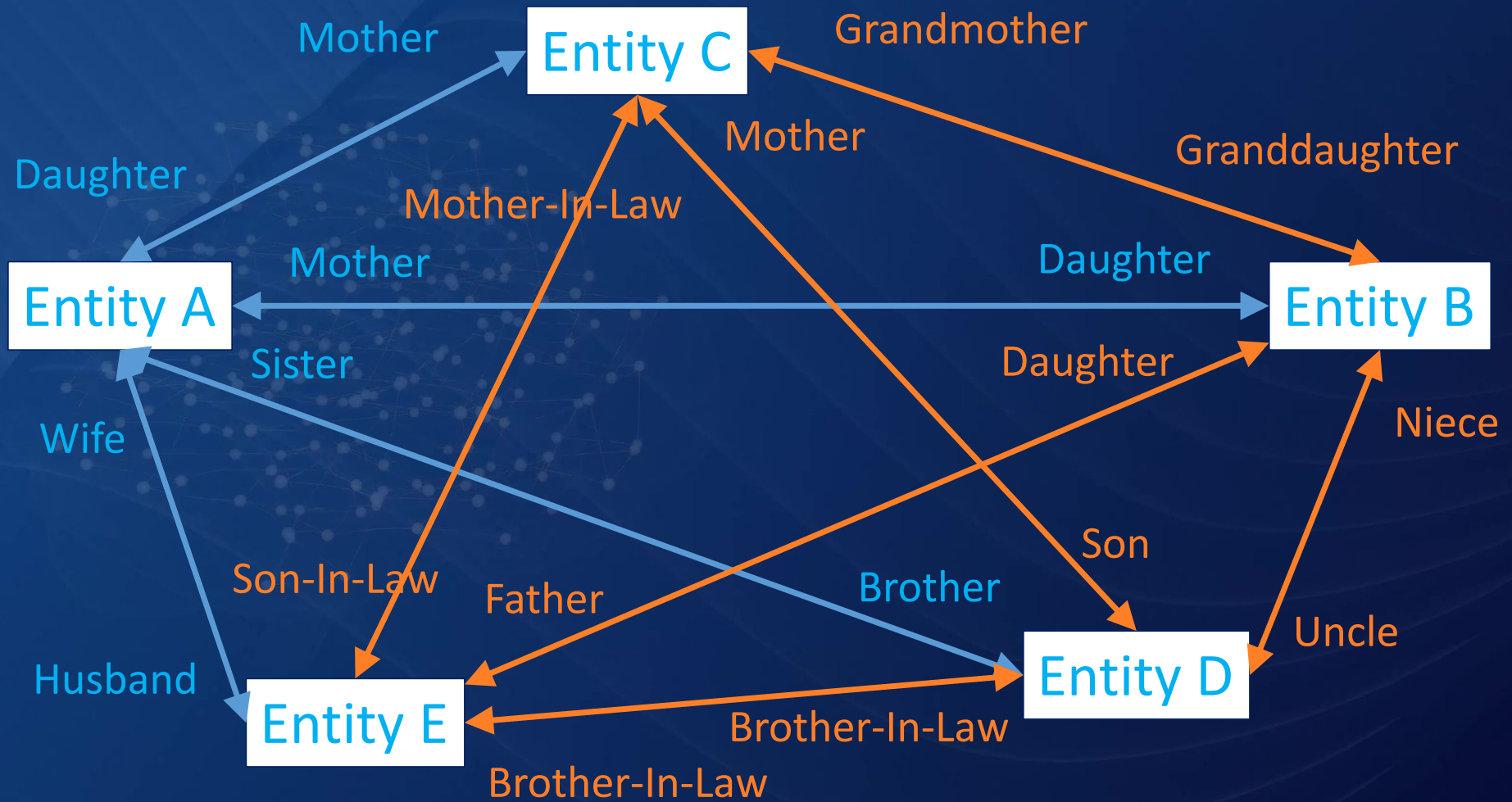




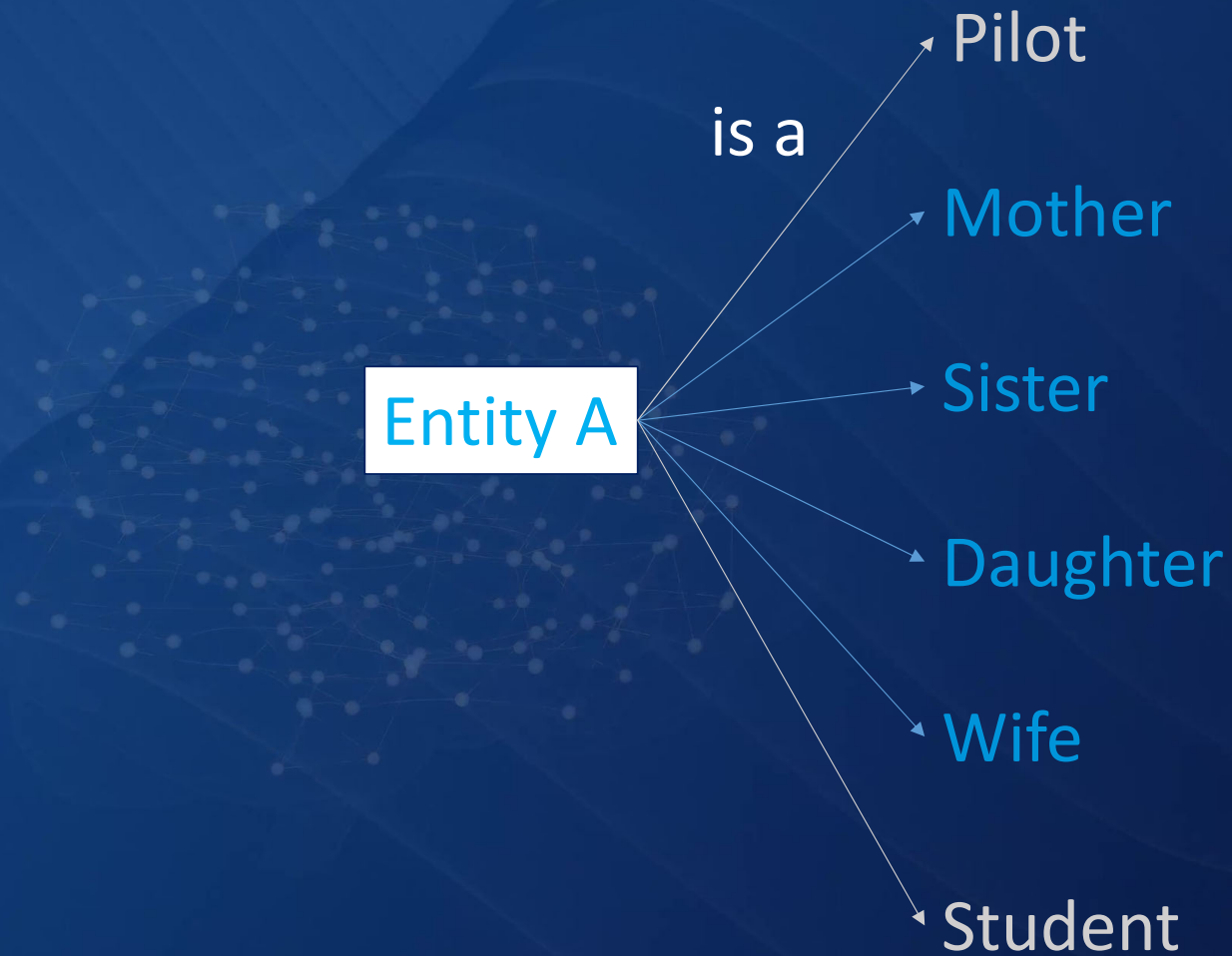
3. Association Network

Foundation for Concepts

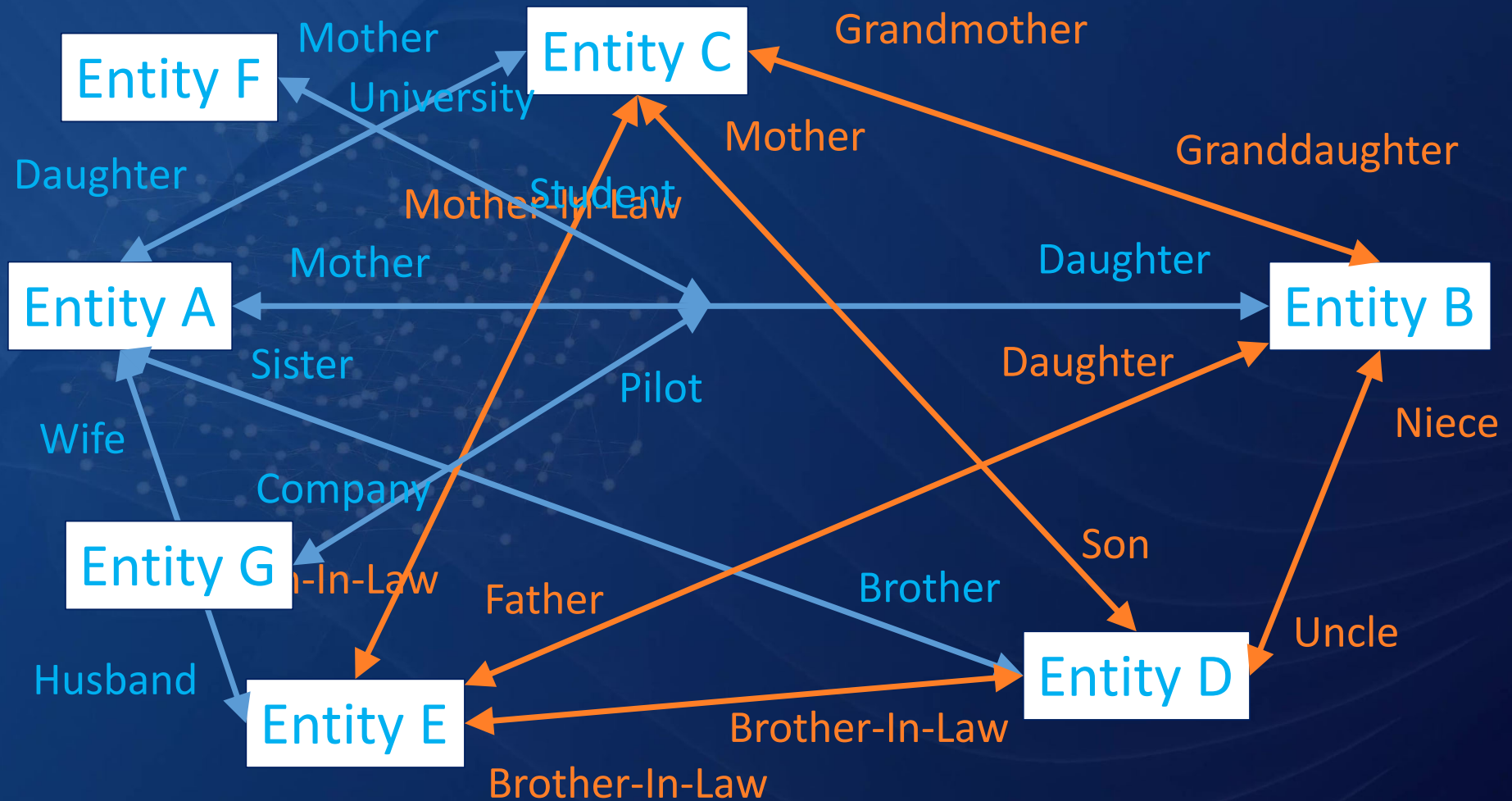
3. Association Network

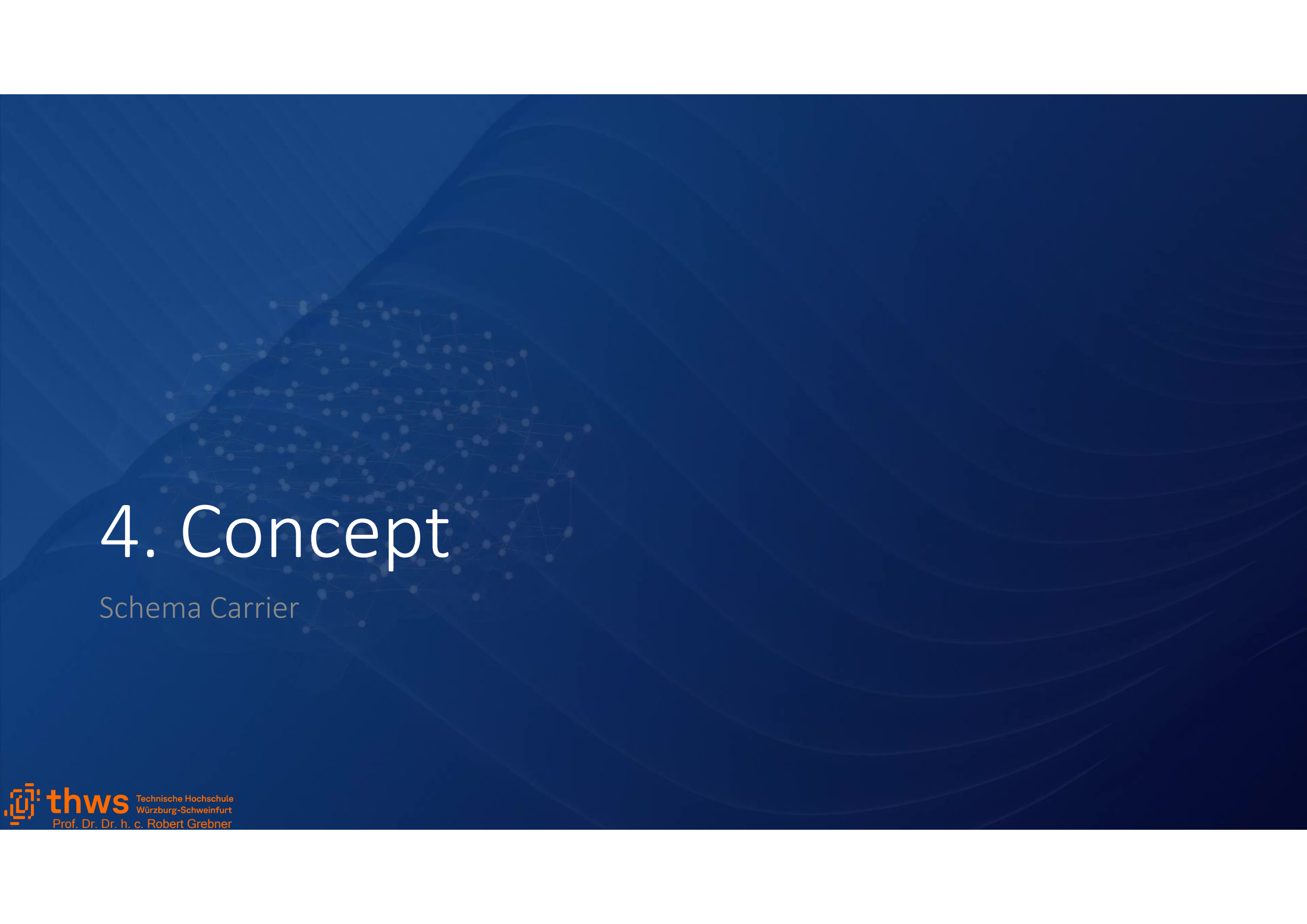


3. Association Network



3. Association Network





4. Concept

Schema Carrier

4. Concept

A **primitive concept** is a mental entity, that stands for it alone, this means that it is not defined by any association schema (e. g. name, ...).

Cornelia

A **native concept** is defined by an association schema and is usually derived from for the physical world (e. g. mother&daughter)



A **derived concept** has a network scheme build of multiple association schemas.

4. Concept

Primitive Concept Name

What is a Name?

Phonetic/Textual Pattern.

When is it Used?

When Individuals
Communicate.

What is it Used for?

To Assign it to Mental
Entities.

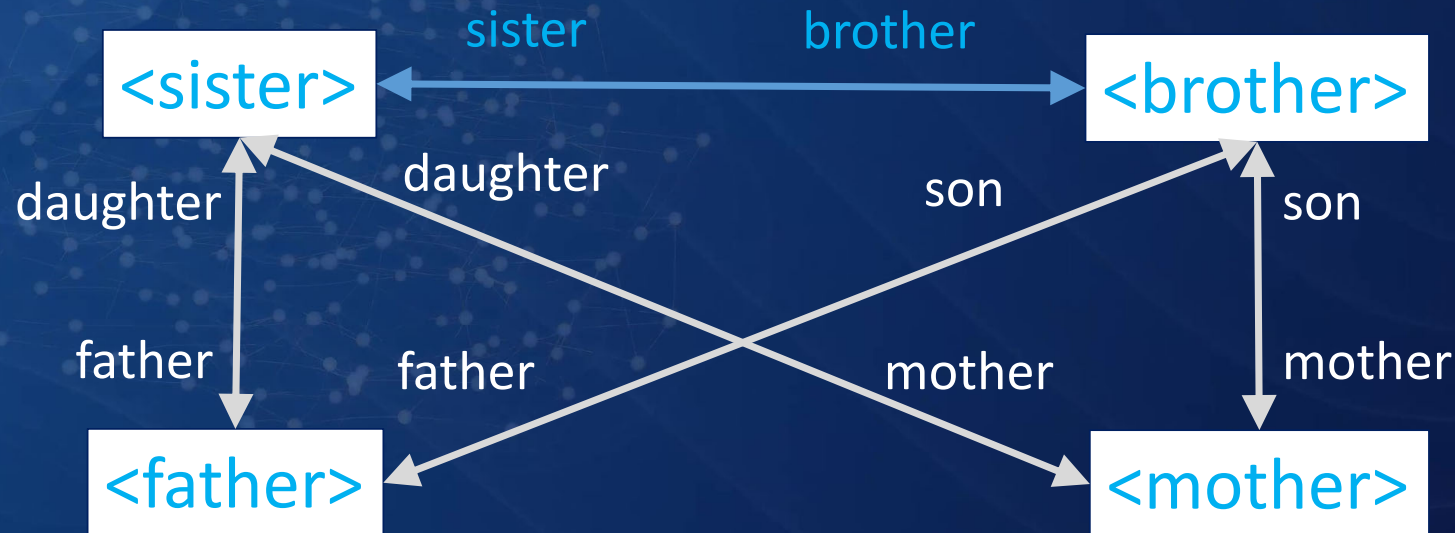
Why is it Used?

To Shortcut the
communication.

4. Concept

Derived Concept

The concept **brother & sister** is derived from the following schema network with the native concepts daughter & mother, daughter & father, son & mother, son & father:

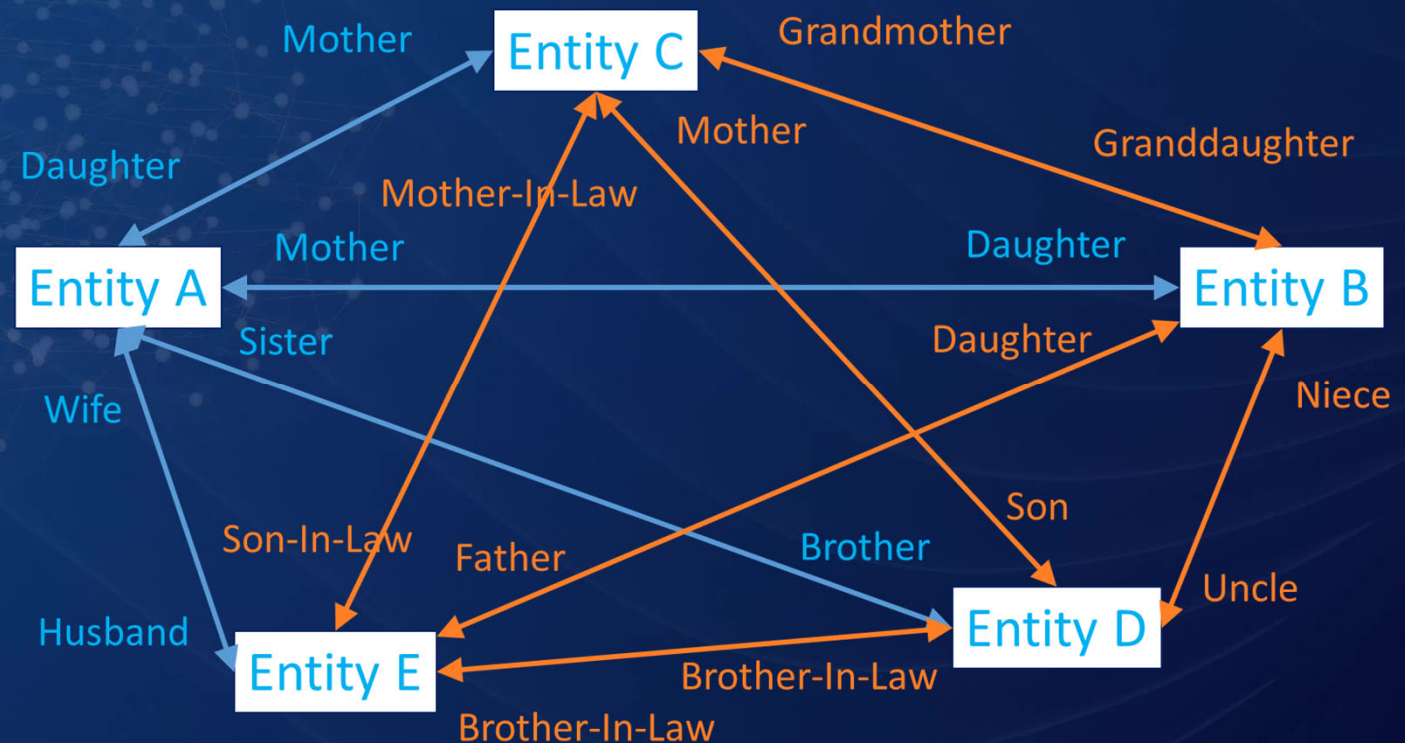


GREBTL (Generic Role-Entity-Based Thought Language):

```
{<sister> (sister & brother) <brother>} (concept & schema) {<sister> (daughter & mother)  
<mother> (mother & son) <brother> (son & father) <father> (father & daughter) <sister>}
```

Workshop „Family“

Find the concept schemas for three of the red drawn non-native concepts in the information network



4. Concept

Why is „hourse not a hourse“?

concept/role $\neq$ physical entity

Physical Entities



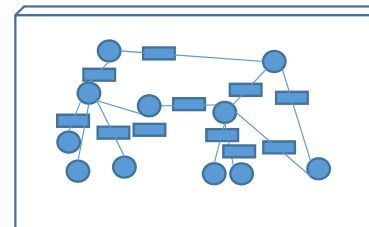
a hourse
(physical
entity)



Mental Entities



hourse
(role)



represents

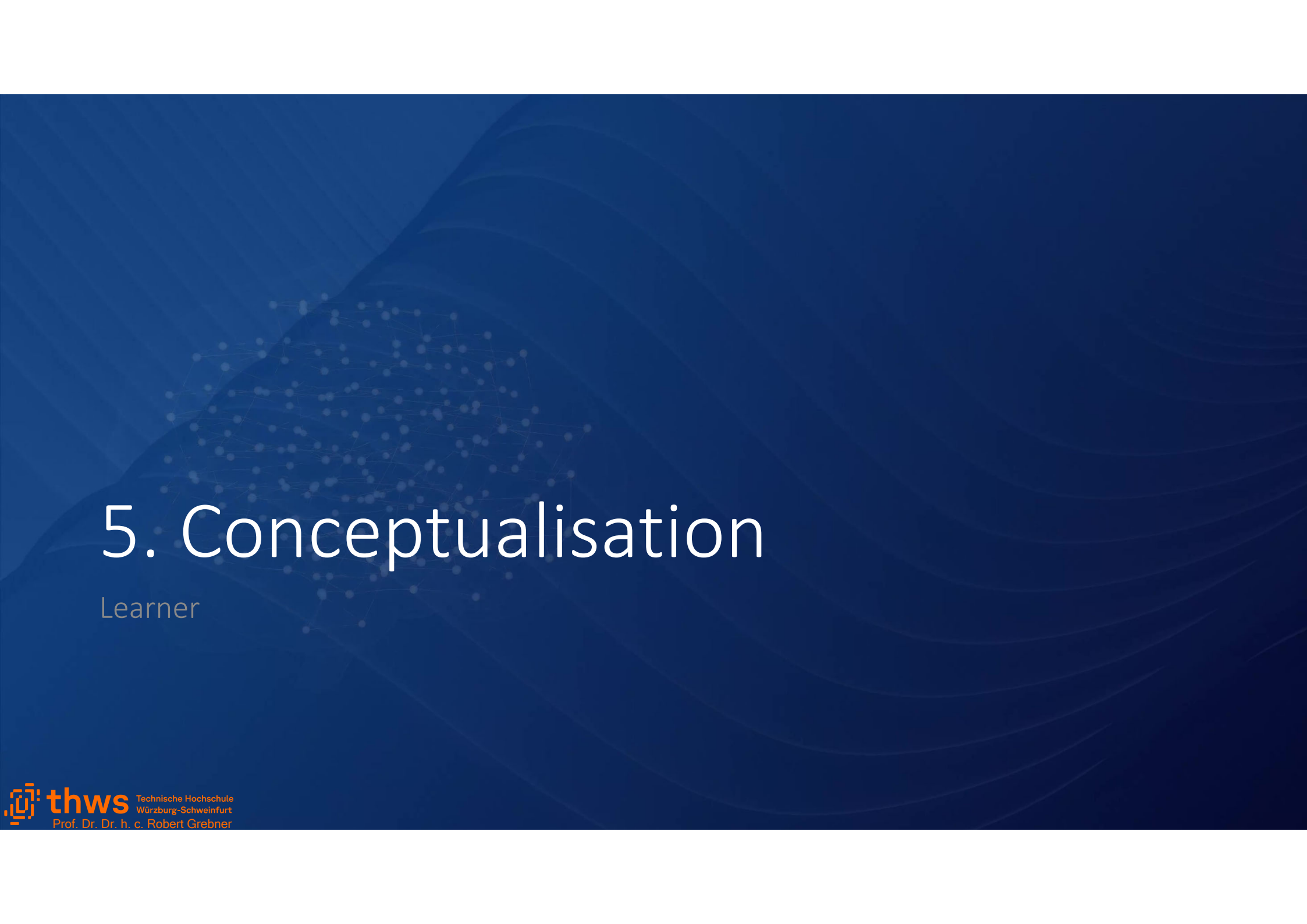
is a

Entity

Workshop „Horse“

Find compatible counterpart roles for the role horse (think about, to what a horse is linked in real life).

Define a „main body part“ concept for a horse which describes the main parts of a horse and their position related to each other.



5. Conceptualisation

Learner

5. Conceptualisation

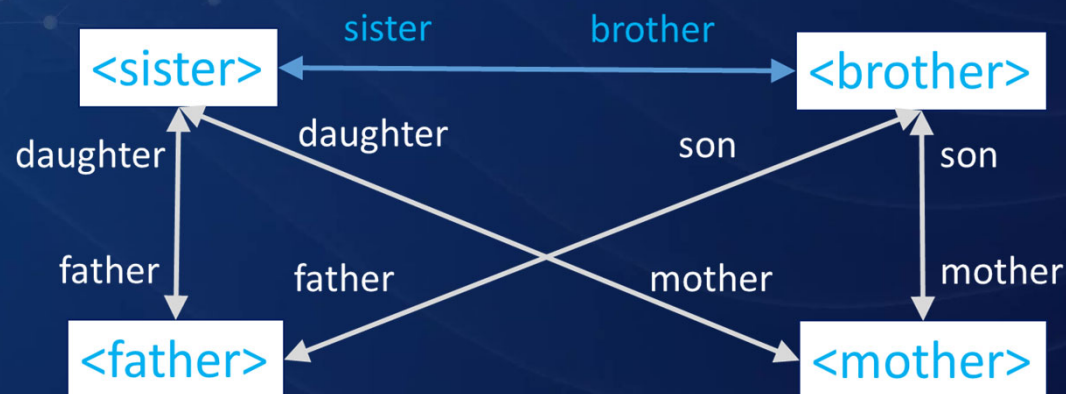
Conceptualisation means to define a concept with its schema

Possible ways to do this is:

- out of experience
- through repeating action patterns
- through combination and/or adaption of known concepts (transfer)
- by other individuals like teachers

Learning means internalising concepts into the mind.

Where did you learn the concept sister&brother?



Workshop „Conceptualisation“

How would you implement a conceptualisation software to get concepts out of received information networks? Think about a system, that should be able to learn the concept sister&brother for example

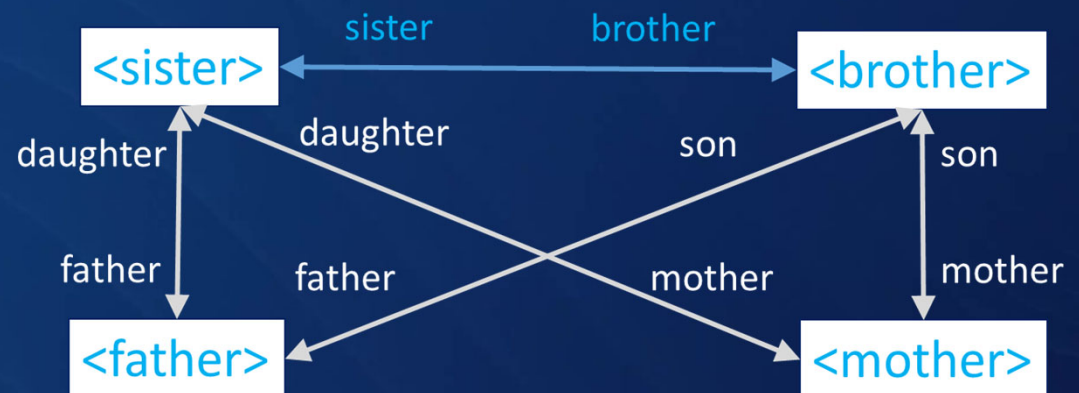
6. Materialisation

Actor

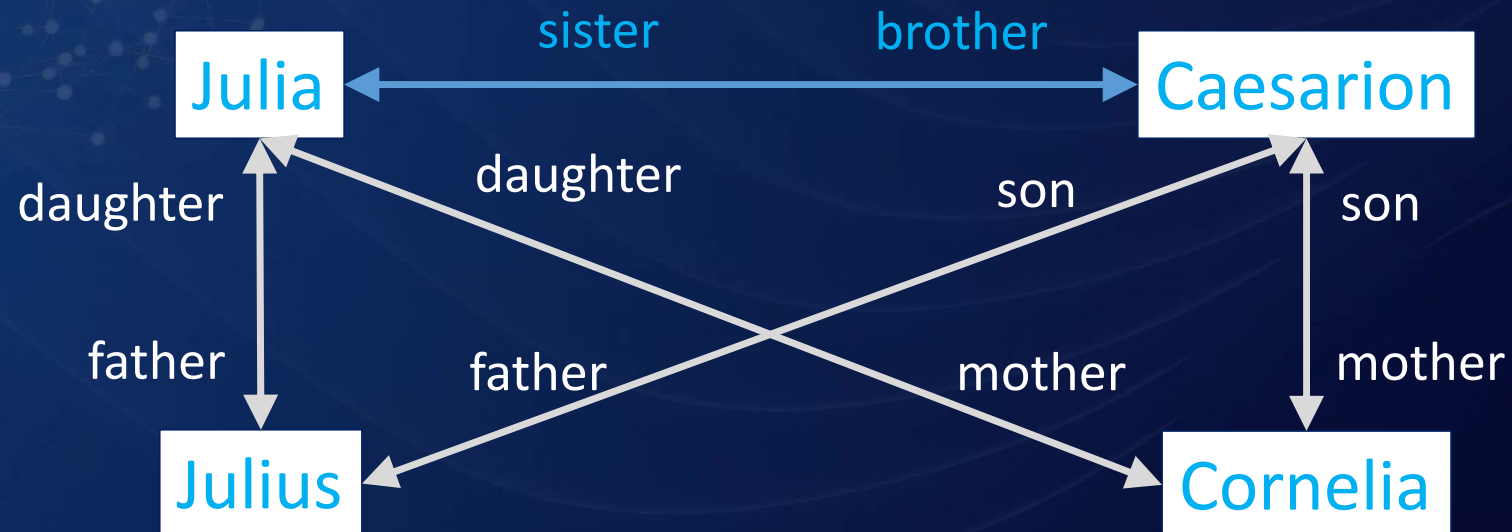
6. Materialisation

Materialisation (instantiation) means to bring a concept into real life by

- Acting (physical world)
- Thinking (mental world)



instantiate
materialise





7. Conclusion

Core and Key

7. Conclusion

- Concepts and their schemas base on association and information networks.
- Nesting is a complex issue and can have different forms:
 - concept is part of an overlaying concept (simple concept)
 - concept is interconnected to other concepts in complex concepts
 - Classical inheritance known from object oriented computer languages
- Concepts are the core of learning and acting.
- Concept schemas are the key to intelligence as well as for curiosity.



I
Thank
You